Internal Specification for Glazed Rectified Porcelain tiles.

Sr #	Characteristics / Size	40x80x2.0 cm		40x80x2.0 cm Die effect	
1	Nominal Size in mm	400.0x800.0x20		400.0x800.0x20	
2	Work size in mm	400.0x800.0x20		400.0x800.0x20	
	Standard	ISO 13006	SCC	ISO 13006	SCC
3	Max Length & width mm	-	399.9 - 400.4 799.9 - 800.4	-	399.9 - 400.4 799.9 - 800.4
4	Max deviation in diagonal mm		0.8		0.8
5	Thickness (mm)		20		20
5.1	Thickness Range mm		19.5 – 20.5		19.50 – 20.50
6	Centre curvature	mm	mm	mm	mm
6.1	a) Convex +	+ 1.80	+ 1.80	+ 1.80	+ 1.80
6.2	b) Concave-	- 1.80	- 0.80	- 1.80	- 0.80
7	Warpage mm	± 1.80	± 1.80	± 1.80	± 1.80
8	Edge curvature mm				
8.1	a) Convex +	+ 1.80	+ 1.40	+ 1.80	+ 1.40
8.2	b) Concave-	- 1.80	- 0.60	- 1.80	- 0.60
9	Max deviation from straightness mm	± 0.8	± 0.50	± 0.8	± 0.50
10	Max deviation from rectangularity mm	± 1.50	± 1.20	± 1.50	± 1.20
11	Group	B1a	B1a	B1a	B1a
12	Water absorption %	≤0.5	≤0.5	≤0.5	≤0.5
13	Breaking strength (N)	≥1300	≥1400	≥1300	≥1400
14	Modulus of rupture N/mm^2	≥35	≥35	≥35	≥35
15	Resistance to surface abrasion class	Manufacturer to state class	3 - 5	Manufacturer to state class	3 - 5
16	Coefficient of linear thermal expansion ($x10^{-6}K^{-1}$)	-	≤ 9	-	≤ 9
17.1	Scratch resistance Moh's scale (Glossy)	-	≥ 3	-	≥ 3
17.2	Scratch resistance Moh's scale (Matt)	-	≥ 6	-	≥ 6
18	Resistance to thermal shock	Required	Resistant	Resistant	Resistant
19	Crazing resistance (5bar / 2hr cycle)	Required	Resistant	Resistant	Resistant
20	Frost Resistance	Required	Resistant	Resistant	Resistant
21	Staining resistance G (min class)	3	3	3	3
22	Chemical resistance to acid & alkalis (min Class)	Required if agreed	B	Required if agreed	B
23	Color resistance to UV light	-	95% tiles without defects	-	95% tiles without defects
24	Resistance to household chemicals (min class)	B	B	B	B